May 09

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1 May 9

1.1 Overview of what I want to do this week

1. Find a solution for the cross-sectionally correlated error terms in the nonlinear beta problem.

2. Read and do exercise on Hall and Heyde, 2014, van der Vaart and Wellner, 1996. And more on measuret theory and functional analysis.

3. Read new papers.

2 May 11

Now I am reading Hardle and Mammen, 1993 and see how we can actually apply the L^2 statistics.

2.1 Hardle and Mammen

Assumption 1. *Some assumptions:*

1. *X_i lies in a compact set in $\mathbb{R}^d$, and marginal density of $X; i > 0$.*

2. *$m(\cdot)$ and $f(\cdot)$ are twice continuously differentiabel, $\pi \in \mathcal{C}^1$.*

3. *$\delta_n(\cdot)$ is uniformly in x, n bounded and $c_n = n^{-1/2}h^{-d/4}$*

4. *$\sigma^2(x)$ is bounded away from $0, \infty$.*

5. *$E\exp\{t\varepsilon_i\}$ is uniformly bounded in i, n for $|t|$ small enough.*

And the assumption on the parametric regression:

$$m_{\widehat{\theta}}(\cdot) - m_{\theta_0}(\cdot) = \frac{1}{n}\sum_{i=1}^{n} \langle g(\cdot), \varepsilon_i h(X_i) | g(\cdot), \varepsilon_i h(X_i)\rangle + o_p\left((n\log n)^{\frac{-1}{2}}\right)$$

$$T_n = nh^{d/2}\int \left(\widehat{m}_n(x) - \mathcal{K}_{h,n} m_{\widehat{\theta}(x)}\right)^2 \pi(x)$$

Then

$$T_n \rightsquigarrow N\left(b_h + \int (\mathcal{K}\delta_n)^2 \pi, V\right)$$

Compare with Li and Racine, 2007 will actually give much resemblence.

2.2 On Central Limit Theorem for U statistic

With dependent random variable, there is Gao and Hong, 2007. I think there are at least three papers that I need to read: Hall, 1984, de Jong, 1987 and the Gao and Hong, 2007. They may all be based on some martingale result.

Kuersteiner and Prucha, 2018 CLT for quadratic terms in a panel settings.

From Kuersteiner and Prucha, 2018 I found the paper Gabaix and Koijen, 2020, which seems very interesting.

3 May 12

Let's first derive the result for L^2 statistic when the error contains the first step estimation error.

I read the chapter 5 and 6 in CLM. I reread the email from Prof. Linton, it seems that actually it contains the basic framework for everything that I have spent a lot of effort to understand.

4 May 13

4.1 Literature Search

One question I am interested in is where we can apply IV method in Finance. It should be useful given that GMM is closely related to IV estimation. If we have a good question, we can use GIV perhaps.

The main theorem of the de Jong, 1987 is that

Theorem 1 (Theorem 2.1 in de Jong, 1987)**.** *If* $W_n = \sum_{i,j} W_{ij}$ *with* $W_{ij} = w_{ij}(X_i, X_j) +$

*$w_{ji}(X_j, X_i)$, X_i are independent and $EW_{ij}^2 = \sigma_{ij}^2 < \infty$ is **clean**, that is*

$$E\left(W_{ij} \mid X_i\right) = 0 \quad a.s.$$

with variance $\sigma(n)^2$, if

1. $\sigma(n)^{-2} \max_i \sum_{j=1}^n \sigma_{ij}^2 \to 0$ *as* $n \to \infty$,
2. $\sigma(n)^{-4} EW(n)^4 \to 3$ *as* $n \to \infty$

$$\sigma(n)^{-1} W(n) \rightsquigarrow N(0,1)$$

This is based on Martingale CLT.

4.2 On nonlinear Beta

Wang, 2003 tests with conditioning information nonparametrically, he has another paper Wang, 2002 has more technical details.

And notice that once he get an error estimate u he tests that $E(u \mid x) = 0$ by running a linear regression of u on all the conditioning variables. He also uses some results about the generalized U-statistics.

I wonder how to put the CAPM in a SDF framework: see Cochrane's online course and does it matter to test the nonlinearity?

So we could put everything in a SDF framework where the pricing kernel is

$$m = (1 - b'f) \quad E\left(mR_i^e\right) = 0$$

In terms of CAPM, let r be excess return of R,

$$E(r_i) = E(r_m) \frac{Er_m r_i}{E(r_m)^2} \tag{1}$$

$$m = (1 - br_m) \quad \text{where} \quad b = \frac{Er_m}{E(r_m)^2}$$
$$E(r_i - br_m r_i) = 0$$

Here b is not the betas but rather the risk premiums. So in terms of linearity we should test whether the function m is a linear function of r_m.

It seems that conditional CAPM tests takes β as a fundamental characteristic of an asset and the risk premium can be changing, something like this:

$$ER_i = \gamma(z)\beta \quad \text{where} \quad z \quad \text{is conditioning variables}$$

but the relationship is still linear in β

For our test it seems that we are allowing:

$$ER_i = g(\beta) \quad \text{where} \quad g' = \gamma$$

But I was wrong. The conditional CAPM allows both β and γ to be conditional. I don't think that Wang, 2003 properly takes care of the conditioning nature of the β's.

I need to practice going forward and back between expected-return-beta model and SDF representation.

5 May 14

5.1 Conditional CAPM doesn't explain the size and value premium

Lewellen and Nagel, 2006 says that the variation in β and risk premium is too small to account for the the pricing errors. And develop a simple *short-window* test for the conditional CAPM and says that this method doesn't need to consider which conditioning information to include.

They did a short-window time series regression test with *portfolios*, and analysed the std.dv like Fama-MacBeth, which should be problematic, especially considering their estimates are noisy.

There is also Freyberger, Neuhierl, and Weber, 2020, which should be relevant.

For the testing with the estimated errors:

1. Generalized liklihood ratio specialized to the regression model case and compare to Hardle and Mammen.

2. Look through the proof in Li and Racine, 2007 and see how to apply the CLT of quadratic forms.

I checked the proof in Fan, C. Zhang, and J. Zhang, 2001 in the varying coefficient models, where the regression function is $A(U)$, we base the result on the RSS_0 and RSS_1 which then rely on the difference $A_0(u) - \widehat{A}(u) = a_n + R_n$, then a lot of approximation takes place, plus the CLT in de Jong, 1987 gives the final result.

6 May 15

We start with trying to find generalization of the de Jong, 1987. Let's read Gao and Hong, 2007.

Notice that in the end the test statistic we might use is

$$\sum_{i,j} \widehat{u}_{it}\widehat{u}_{jt} K_h\left(\widehat{\beta}_i - \widehat{\beta}_j\right)$$

I did some calculation of the test statistic

$$\sum_{i \neq j} \widehat{u}_i \widehat{u}_j$$

based on the result when we do full cross-sectional regression.

$$\widehat{u}_i = \bar{\varepsilon}_i - \widehat{\beta}_i \widehat{A}_\beta \bar{\varepsilon} - \left\{\left(\widehat{\beta}_i - \beta_i\right)\bar{\gamma} - \widehat{\beta}_i \widehat{A}_\beta \left(\widehat{\beta} - \beta\right)\bar{\gamma}\right\}$$

6.1 Time Varying Asset Pricing model

Gagliardini, Ossola, and Scaillet, 2016 seems to answer most of my questions in a parametric way. Important.

There are some features of the model: $i \in [0, 1]$ and we have discrete time, there is also random sampling from $[0, 1]$ but I guess it's not essential. Then they develop a model with linear factor structure, derive testable implication, considered estimation, confidence interval and specification test, which is still based on the squared distance measure. It's a rather complete paper.

7 May 16

7.1 On regression with cross-sectional correlation

Andrews, 2005 which is referenced in Gagliardini, Ossola, and Scaillet, 2016 considers regression model with cross-sectional correlation due to common shocks. He assumes that the error term u_i in a cross-sectional regression are i.i.d conditional on some unobserved common shock C.

It can accommodate arbitrary type of dependence provided that the sampling scheme is random. Can we generalize it to the specification test settings?

There are more nonparametric testing in panel data model. For example, Lin, Li, and Sun, 2014, but they only allow time sereis correlation and they are applied to the whole panel, not the second step.

7.2 Some other readings

On the new Econometrica, there is a paper about social learning equlibria, it's not applicable to financial trading, but I think a description of the financial market using the Game Theory approac to account for the sallient features:

1. Asymmetric behaviour: leverage effect.
2. GARCH type clustering of volatitliy
3. Randomness in return , but with fat tail.
4. Bubble and crashes.
5. Granularity. Power distribution.

Ludvigson has a chapter in the Handbook of Economics for Finance, which mentions heterogenous agents. Might worth a look.

7.3 Gabaix and Koijen GIV

It seems that any weights that are not uniform will get rid of the common component, is there a better way to construct the IV? How does it relate to nonparametric IV models. It allows for screening using news and economic sense, can we make it scientific.

In section 2, without a loop, can construct given demand disturbance y_{it} in $D_{it} = QS_i(1 + y_{it})$ and S_i size, construct instrument

$$z_t = \sum S_i y_{it} - \sum \frac{1}{N} y_{it}$$

is a valid instrument because it get rid of the systematic/common component in y that might be correlated to the shock ε_t to the pricing equation:

$$p_t = \alpha y_{St} + \varepsilon_t$$

Then it talks about classical estimation of demand and supply elasticity using GIV.

Section 2.5 talks about the possibility that the idiosyncratic part of the shock, that is u_{it} in $\lambda_i \eta_t + u_{it}$, will actually contain a part that comes from shock to the loading λ of the common shock η.

In section 5, they discuss a spillver model of sovereign debt.